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Exogenous citrate restores the leaf metabolic profiles of navel orange plants under boron deficiency

Guidong Liu [a,b,*], Tian Jin [a], Yuemei Xu [a], Fengxian Yao [a,b], Guan Guan [a,b], Gaofeng Zhou [a,b]

[a] College of Life Sciences, Gannan Normal University, Ganzhou, Jiangxi Province, 341000, PR China
[b] National Navel Orange Engineering Research Center, Ganzhou, Jiangxi Province, 341000, PR China

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ABSTRACT

Boron (B) is an essential micronutrient for higher plants, and its deficiency causes a change in the citrate concentration in leaves of young navel orange plants. Although citrate has been implicated in the regulation of gene expression for many transcripts, it is unclear whether citrate can affect metabolic profiling under B deficiency and if so, how many metabolites are affected. In this study, GC-TOF-MS-based untargeted metabolite profiling was used to identify the physiological effects of exogenous citrate on recovery of metabolites in B-deficient orange plants. There were 31 increased and 24 decreased metabolites in the boron-deficient (BD) group leaves relative to those of the boron-adequate (BA) group. Boron deficiency-induced changes in many metabolites were restored to the level of BA (control) group leaves or showed a recovery tendency at 1 week after citrate supply (foliar application of citrate, BDFC), including 11 organic acids, 9 sugars and polyols, 10 amino acids, and 4 other compounds. To compare with the metabolic recovery effects of exogenous citrate on B deficiency, exogenous application of B (borate) was also performed under same conditions (BDFB), and similar effects on the regulation of metabolic homeostasis under B deficiency were observed. Both the results of principal component analysis (PCA) and hierarchical cluster analysis (HCA) showed that BA, BDFC, and BDFB were relatively similar and clustered close to each other. There are different responsive and regulatory mechanisms to the additions of exogenous citrate in navel orange leaves under B adequate and deficient conditions. Based on these results, we suggest that citrate is an important component of the B deficiency stress response, and exogenous application of citrate generally restores the leaf metabolic profiles of navel orange plants under boron deficiency, which might play a positive role in this stress tolerance.

1. Introduction

Boron (B) is an essential micronutrient for higher plants, and its deficiency causes various metabolic disorders and growth defects in many agricultural crops, including citrus plants (Shorrocks, 1997). The role of B as a structural element in plant cell walls and membranes has been well investigated (O'Neill et al., 2001; Voxeur and Fry, 2014). Boron deficiency results in the alteration of the network of pectin and cell wall components and structure, which induces the visible symptoms including curling of the leaves and significant enlargement of root tips in citrus plants (Liu et al., 2014; Wu et al., 2020). Compared to the demonstrated structural role of B as an assembly element in the cell wall and membrane, involvement of B in the functioning of plant cell metabolism is still a subject of considerable debate (Brown et al., 2002). While changes in the levels of several plant metabolites such as sugars, amino acids, and phenolics following B deficiency stress are extensively documented (Han et al., 2008; Alves et al., 2011; Liu et al., 2015), knowledge of changes in a certain metabolite and direct or indirect effects of B stress and their underlying mechanism is still incomplete (Chen et al., 2014).

Citrate is widely acknowledged to play a central role in plant metabolism (Sweetlove et al., 2010). Citrate is involved in the production of energy mainly through the tricarboxylic acid (TCA) cycle in mitochondria, and its storage also contributes to the cellular osmolyte pool and participates in regulating cellular pH (López-Bucio et al., 2000). The exudation of citrate has been reported to be closely related to the stress tolerance of heavy metals, phosphorus deficiency, alkaline stress, and high salinity (Panchal et al., 2021). Exogenous citrate could

* Corresponding author. College of Life Sciences, Gannan Normal University, Ganzhou, Jiangxi Province, 341000, PR China.
E-mail addresses: liuguidong@gnnu.edu.cn (G. Liu), 736065589@qq.com (T. Jin), 2646907824@qq.com (Y. Xu), fengxianyao@aliyun.com (F. Yao), guanguan_1985@aliyun.com (G. Guan), zhougaofeng428@163.com (G. Zhou).

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also improve stress responses and tolerance. It significantly improved plant growth and the internal citric acid concentration and also induced defense mechanisms by increasing the activities of antioxidant enzymes (Sun and Hong, 2011; Hu et al., 2016). In addition, it has been confirmed that the citrate effect is not only due to pH changes, and there are a variety of physiological responses to applied citrate (Ghazijahani et al., 2014; Massoud et al., 2019). However, little information has been reported about citrate-related stress defense mechanisms on plant growth and metabolic regulation in plants, especially with regard to exogenous citrate, until now.

In our previous study, we found that B deficiency decreased the citrate concentration in leaves of young navel orange plants (Liu et al., 2015). As a key metabolite signaling molecule, citrate has been shown to act as a regulator of gene expression in *Arabidopsis thaliana* (Finkemeier et al., 2013). Specific functional categories of transcripts that are regulated by citrate include the TCA cycle, nitrogen metabolism, sulfur metabolism, and DNA synthesis (Finkemeier et al., 2013). Therefore, it is possible that the changes in some metabolites induced by B deficiency may be caused by a decreased level of citrate rather than a direct effect of B deficiency. Here, we hypothesize that exogenous citrate may partially restore the levels of metabolites that are changed due to B deficiency. Although some carboxylic acids have been implicated in the regulation of gene expression for many transcripts (Finkemeier et al., 2013), it is unclear whether citrate could affect metabolic profiling under B deficiency and if so, how many metabolites are affected.

So far, there is limited information on the effects of foliar citrate on cellular metabolism under normal and B deficiency conditions. Therefore, the aim of this study was to investigate the physiological effects of exogenous citrate on the recovery of metabolites in B-deficient orange plants. We also compared the effect of the foliar application of citrate and borate on metabolic profiling in navel orange plants exposed to B deficiency solution. This study provides new insights into the metabolic mechanism of citrate in navel orange response to B deficiency.

2. Materials and methods

2.1. Plant material and treatments

One-year-old navel orange plants (*Citrus sinensis* cv. Newhall) grafted on trifoliate orange (*Poncirus trifoliate* (L.) Raf.) rootstocks were used. All plants were uniform in stem diameter (6–8 mm) and height (30–35 cm). They were severely pruned in order to develop new roots, stems and leaves, in response to B treatment. All lateral shoots of the scion and the major part of the root system were cut, while the main shoot of the scion was defoliated and decapitated 10 cm above the grafting zone (Papadakis et al., 2004). Subsequently, plants were washed with distilled water and transplanted into black plastic bags containing 8 L of a B-free sand:perlite (1:1, v/v) mixture. After transplanting, plants were transferred to a greenhouse with sunlight (relative humidity of 60%–80% and temperature of 25–32 °C) and were irrigated for 1 month with good quality tap water, until new shoots appeared. At the initiation of the experiment, only two growing shoots (7–10 cm height, the leaves were not fully expanded) having almost equal leaf area were maintained on each plant. The treatments started at the beginning of May 2019 and lasted 4 weeks. During this period and every 2 days, plants were irrigated with a modified Hoagland's No. 2 nutrient solution (Hoagland and Arnon, 1950) with (20 μM H_3BO_3, as BA treatment) and without (no B added, as BD treatment) boron. The composition and salt contents of the nutrient solution were as follows: 3 mM KNO_3, 2 mM $Ca(NO_3)_2$, 1 mM $MgSO_4$, 0.28 mM Na_2HPO_4, 0.64 mM NaH_2PO_4, 9 μM $MnCl_2$, 1.6 μM $ZnSO_4$, 0.32 μM $CuSO_4$, 0.2 μM Na_2MoO_4, and 50 μM EDTA-Fe. To prevent salt and B accumulation in the growth medium, the plants were irrigated with 10 L of deionized water once a week, followed by application of 3 L nutrient solution, to ensure some excess solution drained out of the pot bottom.

After 3 weeks of nutrient solution supply with and without B treatments, all the leaves (fully expanded) of each plant in the exogenous citrate treatment were immersed intermittently, i.e., 20 s every 30 min, a total of 60 s, in 10-mM citrate. Half of the BA and BD plants were treated with deionized water as a control. Additionally, an exogenous boron treatment (all the leaves were immersed intermittently, i.e., 20 s every 30 min, a total of 60 s, in 30 mM borate) was conducted on the BD plants. After 1 continuous week of foliar citrate or borate, the leaves of each treatment were harvested. The experimental design is shown in Fig. 1. The experiment was carried out with a completely randomized design with 5 treatments, i.e., BA, BAFC, BD, BDFC, and BDFB. Each treatment had 3 independent plants.

2.2. Sampling and boron analysis

At the end of the experiment, all plants were harvested, rinsed in deionized water, and then divided into separate parts (roots, stems and leaves). All the leaves of each treatment were further divided into two equal parts; one part (dried to constant weight in the oven) was used for the determination of B concentration, and the other part (grinded into powder with liquid nitrogen) was stored at −80 °C and used for the extraction of metabolites. To measure the B concentration, finely powdered plant samples were dry-ashed at 500 °C for 5 h, after which the ash was dissolved in 10 ml 5% HNO_3, and the B concentration was measured by ICP-MS (Agilent 7900, USA).

2.3. Determination of citrate concentration in leaves

Frozen leaf samples (0.2 g) were homogenized with 2 mL deionized water, followed by ultrasound treatment for 30 min while incubating in ice water. The homogenate was centrifuged (4 °C, 8000 g) for 20 min, and then the supernatant was collected and used for citrate assay. Citrate was quantified using high-performance liquid chromatography (HPLC, LC-100) with a C-18 column (250 × 4.6 mm, 5 μm). HPLC elution buffer was CH_3OH:NaH_2PO_4 (5:95, v/v). The injection volume was 10 μL and the flow rate was 0.8 mL min^{-1}. Chromatograms were run for 30 min using a detection wavelength of 210 nm.

2.4. GC–MS-based metabolomics analysis

Untargeted metabolomics analyses were carried out according to the following steps. (i) *Metabolites extraction*: In brief, 50 mg of leaf (grinding into powder with liquid nitrogen) was extracted using 450 μL extraction

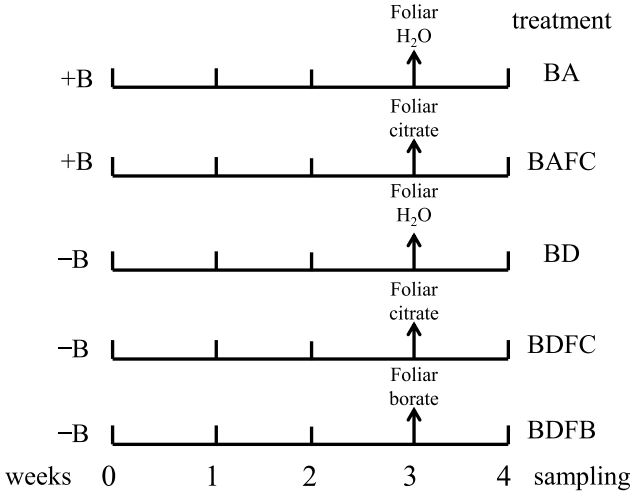


Fig. 1. Diagram showing the boron supply in the solution and foliar application of citrate or borate. BA: boron adequate—the control; BAFC: BA + foliar citrate; BD: boron deficient; BDFC: BD + foliar citrate; BDFB: BD + foliar borate.

reagent (methanol and chloroform, 3:1, v/v) with 10 μL of L-2-Chlorophenylalanine (1 mg/mL stock in dH_2O) as internal standard in a 2 mL centrifuge tube. The samples were then mixed for 30 s and homogenized in ball mill for 4 min at 45 Hz, followed by ultrasound treatment for 5 min while incubating in ice water. Subsequently, the tubes were centrifuged at 13,800 g at 4 °C for 15 min. Supernatant (200 μL) was transferred to a new 1.5 mL EP tubes, and 50 μL was taken from each sample and pooled as a quality control (QC) sample. (ii) *Sample derivatization*: Samples were dried in a vacuum concentrator without heating after adding 80 μL of methoxyamine hydrochloride (20 mg/mL in pyridine), and then incubated for 30 min at 80 °C. BSTFA [Bis(trimethylsilyl)trifluoroacetamide] reagent (1% TMCS: Trimethylchlorosilane, v/v) was added (100 μL) to the sample aliquots, and the QC samples were then incubated at 70 °C for 1.5 h. Then, 5 μL of FAMEs (Fatty Acid Methyl Ester) was added to samples, and they were cooled to room temperature and mixed thoroughly for GC-MS analysis. (iii) *GC-TOF-MS analysis*: GC-TOF-MS analysis was performed using an Agilent 7890 gas chromatograph system coupled with a Pegasus HT time-of-flight mass spectrometer. A 1 μL aliquot of the analyte was injected in splitless mode. Helium was used as the carrier gas, the front inlet purge flow was 3 mL min^{-1}, and the gas flow rate through the column was 1 mL min^{-1}. The initial temperature was kept at 50 °C for 1 min, then raised to 310 °C at a rate of 10 °C min^{-1}, then kept for 8 min at 310 °C. The injection, transfer line, and ion source temperatures were 280, 280, and 250 °C, respectively. The energy was −70eV in electron impact mode. The mass spectrometry data were acquired in full-scan mode with the m/z range of 50–500 at a rate of 12.5 spectra per second after a solvent delay of 6.33 min. (iv) *Data processing*: Chroma TOF 4.3X software of LECO Corporation and LECO-Fiehn Rtx5 database were used for raw peaks exacting, the data baselines filtering and calibration of the baseline, peak alignment, deconvolution analysis, peak identification and integration of the peak area (Kind et al., 2009). Both of mass spectrum match and retention index match were considered in metabolites identification. Remove peaks detected in <50% of QC samples or RSD > 30% in QC samples (Dunn et al., 2011). Peaks with a similarity index of more than 70% and less than 70% were tentatively identified as metabolites and unknown metabolites, respectively (Storey and Tibshirani, 2003). The metabolic data analysis was performed using the internal standard normalization method, with the peak area of each compound being integrated and the peak area ratio of each compound to a corresponding internal standard calculated as the response.

2.5. Statistical analysis

The web-based analytical pipeline MetaboAnalyst 4.0 was used for metabolomics data analysis including data normalization and statistical analysis (Chong et al., 2018). The data for principal component analysis (PCA) were previously auto-scaled (mean-centred and divided by the standard deviation of each variable) to reduce the scale variability among groups, which would thus increase reliability of the results obtained (Xia and Wishart, 2011). The heat-map was constructed based on the putatively identified metabolites to provide a global perspective of metabolite changes in the leaves under different treatments. To understand which metabolites carry the class separating information, partial least squares discriminant analysis (PLS-DA) was conducted, and VIP (variable importance in projection) values were generated. The Kyoto Encyclopedia of Genes and Genomes (KEGG, http://www.genome.jp/kegg/) was used to search for the pathways of the metabolites. MetaboAnalyst 4.0 software was employed for pathway analysis and visualization.

All data (except the metabolomic data) are expressed as means ± standard deviations. Significant differences in B concentrations between the treatments were determined using one-way ANOVA and the Tukey test ($P < 0.05$), while the differences in the metabolites between two groups were determined using Student's t-test ($P < 0.05$).

3. Results

Leaf tissues were collected from control (BA) and B-deficient (BD) plants at 4 weeks following foliar application of citrate or borate treatment. Leaves did not show any visible symptoms in all plants, while BD treatment decreased root growth (data not shown).

3.1. Boron and citrate concentrations in the leaves of different treatments

Significantly lower B concentrations in leaves were detected in foliar citrate-treated plants compared to untreated plants under B-adequate supply conditions (BAFC vs. BA). In contrast, the exogenous citrate application did not significantly affect the B concentration of leaves under B deficiency (BDFC vs. BD). The foliar application of borate induced significant increases in B concentration under B deficiency (BDFB vs. BD). In addition, the B concentration of the leaves in the plants of adequate B supply in solution was much higher than that in the B-deficient plants, regardless of whether the leaves were treated with citrate or borate (Fig. 2a).

Boron deficiency significantly decreased the citrate concentration of leaves in orange plants. The foliar application of citrate or borate both induced significant increases in citrate concentration under B deficiency (BDFC vs. BD and BDFB vs. BD). In addition, significantly higher citrate concentrations in leaves were also detected in foliar citrate-treated plants compared to untreated plants under B-adequate supply conditions (BAFC vs. BA) (Fig. 2b).

3.2. Metabolomic analysis of the leaves in different treatments

Untargeted metabolite profiling was used to determine metabolic differences between citrus leaves of different treatments. The total ion current chromatograms from GC-TOF-MS analysis are shown for the 30 samples in Fig. A.1. A total of 658 peaks were extracted and assigned to compounds (including internal standard) between different sample groups based on the obvious chromatographic differences. Unsupervised principal component analysis (PCA) was employed to evaluate the variations of the metabolic levels of orange leaves under different treatments. As shown in the score plot (Fig. 3), the samples from the BD and BAFC plants were separated clearly from those of the other treatments; however, the samples from the BDFC, BDFB, and BA plants obviously overlapped. The heat-map and hierarchical cluster analysis generated using the concentrations of identified compounds (64 metabolites) in 30 leaf samples (i.e., five treatments and six replicates) showed that BA, BDFC, and BDFB were relatively similar and clustered close to each other (Fig. 4).

3.3. Overall changes in metabolites and metabolic pathway in response to B deficiency

PCA showed an obvious separation between the BD and control treatments, indicating that the B-deficient leaves exhibited changes in their metabolic profiles relative to the control leaves (Fig. A.2). On the basis of the Student's *t*-test results, 55 of 64 identified metabolites in leaves changed significantly under B deficiency stress (31 up-regulated and 24 down-regulated metabolites) compared to the control, including 18 amino acids, 14 sugars and polyols, 16 organic acids, and 7 other compounds (Table A.1). The significantly different metabolites induced by B deficiency in the central metabolic pathways were shown in Fig. 5 as a metabolic network based on the KEGG (Kyoto Encyclopedia of Genes and Genomes) pathway database.

To explore the potential metabolic pathways that were affected by B deficiency, the differential metabolites (Table A.1) were imported into MetaboAnalyst 4.0. The pathway analysis results provided a detailed account of the metabolic pathway changes associated with B deficiency (Fig. 6). On the basis of both –log(p)-values > 15 and pathway impact scores >0.2, the most relevant metabolic pathways were identified as

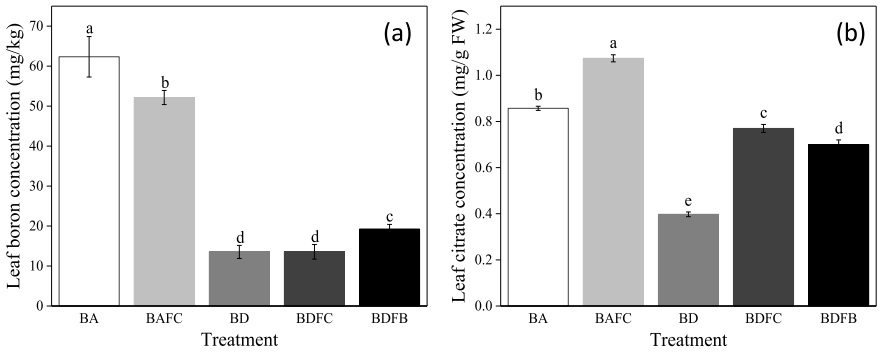


Fig. 2. Boron concentration (on a dry weight basis, a) and citrate concentration (on a fresh weight basis, b) in the leaves of different treatments. *Bars* are the means of three replicates ± SD. Different letters in each group indicate significant differences ($P < 0.05$).

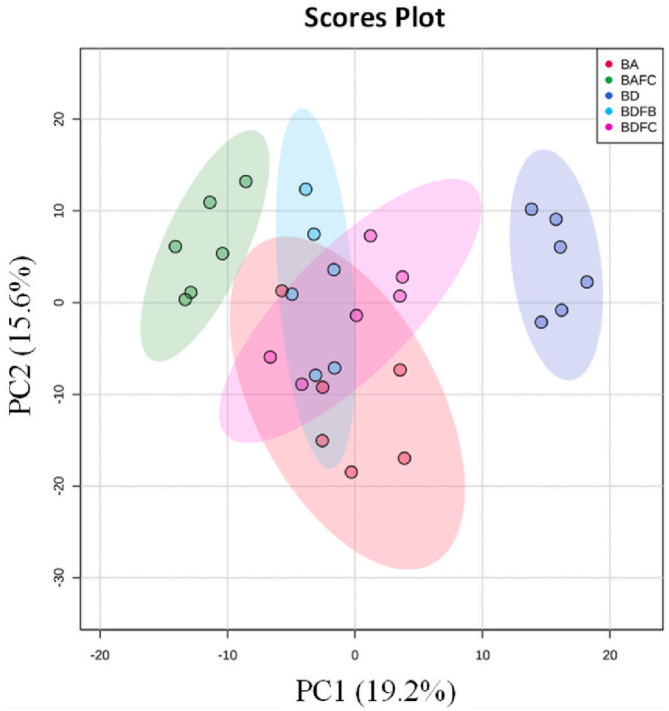


Fig. 3. Principal component analysis (PCA) score plot of GC-MS metabolomic profiles in leaves of navel orange plants. PCA and the graphic representation were carried out using the MetaboAnalyst platform.

pyruvate metabolism, citrate cycle (TCA cycle), glyoxylate and dicarboxylate metabolism, and glycine, serine and threonine metabolism (Fig. 6).

3.4. Metabolomics affected by exogenous citrate under B adequate and deficient conditions

Partial least squares-discriminate analysis (PLS-DA) was used to evaluate metabolic profile changes. In particular, pairwise comparisons were performed for the BAFC vs. BA and BDFC vs. BD groups to evaluate the exogenous citrate effect on metabolites under normal and B deficiency conditions. Clear separations between groups were observed in the PLS-DA score plots (Fig. 7a and b), indicating marked differences in the metabolic characteristics of the groups.

Some metabolites with VIP (the variable importance in projection) scores >1 in the PLS-DA model play an important role in separating groups. Based on an evaluation of the relative metabolic changes observed in each plant group, the PLS-DA provided 31 and 36 metabolites for the BAFC vs. BA and BDFC vs. BD groups, respectively. For the effect of citrate under adequate B supply (BAFC vs. BA), the top 20 metabolites in leaves contributing to component 1 (accounting for approximately 51.5%) were tyrosine, saccharic acid, glycine, gentiobiose, valine, and 15 other metabolites (Fig. 7c and Table A.2). In contrast, for the effect of citrate under B deficiency (BDFC vs. BD), the top 20 metabolites in leaves contributing to component 1 (accounting for approximately 49.8%) were L-malic acid, cellobiose, gluconic acid, valine, galactinol, and 15 other metabolites (Fig. 7d and Table A.2).

On the basis of both –log(p)-values > 15 and pathway impact scores >0.2, a total of 3 differential metabolite pathways were detected in the BAFC and BA comparison. These pathways were isoquinoline alkaloid biosynthesis, glyoxylate and dicarboxylate metabolism, and the citrate cycle (TCA cycle) (Fig. 8a). In contrast, pathway analysis indicated that only pyruvate metabolism most strongly segregated the BDFC and BD group profiles and was the most statistically significant (Fig. 8b). Taken together, these results suggest that the exogenous citrate effect on metabolite profiles was dependent on the boron supply.

3.5. Recovery of metabolites in leaves of the B-deficient treatment by exogenous citrate

Foliar application of citrate was analyzed in navel orange leaves to understand the metabolomic responses under B deficiency conditions. There were 31 increased and 24 decreased metabolites in the BD group leaves relative to those of the BA group (Table S1). Boron deficiency-induced changes in many metabolites were restored to the level of BA (control) leaves or showed a recovery tendency 1 week after citrate supply (foliar application of citrate, BDFC) (Fig. 9), including 11 organic acids (lactic acid, L-malic acid, pyruvic acid, citric acid, citramalic acid, quinic acid, stearic acid, ketoglutaric acid, shikimic acid, ferulic acid, palmitic acid), 9 sugars and polyols (cellobiose, galactinol, myo-inositol, raffinose, gluconic acid, xylose, glucose-1-phosphate, D-galacturonic acid, galactonic acid), 10 amino acids (oxoproline, tyrosine, glutamic acid, valine, alanine, isoleucine, GABA, serine, glycine, 3-cyanoalanine), and 4 other compounds (phytol, N-Acetyl-beta-D-mannosamine, ethanolamine, glycerol).

To compare with the metabolic recovery effects of exogenous citrate on B deficiency, exogenous application of B (borate) was also performed under the same conditions, and similar effects on the regulation of metabolic homeostasis under B deficiency were observed. After foliar application of borate under B deficiency conditions, 37 metabolites were restored to the level of the BA (CK) group leaves or showed a recovery tendency (Fig. 9). Among these, there were 28 metabolites with a similar recovery tendency between BDFB and BDFC group leaves relative to the BD leaves (Fig. 9), including 10 organic acids (citric acid, L-malic acid, pyruvic acid, lactic acid, quinic acid, shikimic acid, ketoglutaric acid, stearic acid, palmitic acid, and ferulic acid), 6 sugars and polyols (cellobiose, raffinose, galactinol, myo-inositol, gluconic acid, and D-

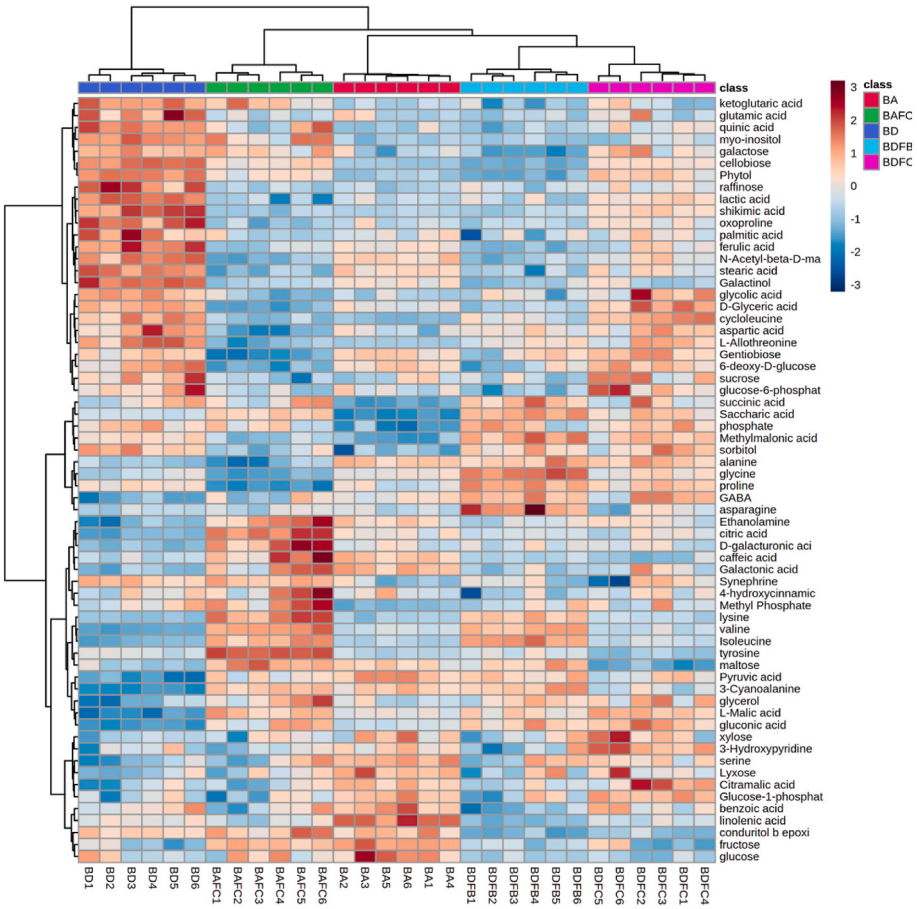


Fig. 4. Heat-map analysis combined with hierarchical cluster analysis (HCA) of the 64 identified metabolites in leaves.

galacturonic acid), 8 amino acids (glutamic acid, valine, alanine, isoleucine, glycine, GABA, oxoproline, and 3-cyanoalanine), and 4 other compounds (phytol, ethanolamine, glycerol, and N-Acetyl-beta-D-mannosamine).

4. Discussion

Untargeted metabolomics, which is a relatively quantitative method, was performed in this study to find more differential metabolites based on GC-MS. The relative concentration means the peak area of each compound to a corresponding internal standard calculated as the response. To verify the accuracy of the present metabolomics data, we determined the actual concentrations of citrate using the HPLC method. The concentrations of citrate obtained from the metabolomics data were consistent with the HPLC method (Figs. 2b and 4, and Tables A.1 and A.2).

4.1. Effect of B deficiency on metabolites in navel orange leaves

It is widely accepted that B deficiency inhibits the growth of plants and affects many metabolic processes such as carbohydrate and phenol metabolism (Dell and Huang, 1997; Brown et al., 2002). In the present study, a total of 55 out of 64 identified metabolites showed a significant change in the concentration due to B deficiency, indicating that B deficiency destroyed metabolic homeostasis in the leaves of navel orange plants. Leaf concentrations of glucose and fructose were notably lower in BD plants than in BA plants. In tobacco roots, a similar decrease in the concentrations of glucose and fructose was detected after 24 h of B deficiency (Beato et al., 2014). Accumulation of secondary metabolites such as phenolics is considered a typical and common effect of B deficiency in plant leaves (Liu et al., 2015). Phenolic metabolism is closely related to carbohydrate metabolism and provides the precursors for the synthesis of phenolic compounds via the shikimate pathway (Payyavula et al., 2014). Compared with controls, B-deficient leaves accumulated higher concentrations of shikimate, quinate, and tyrosine, suggesting that the carbon flux shifted towards the shikimate pathway. Similar results were also observed in trifoliate orange roots (Wu et al., 2020). Shikimate is synthesized through phosphoenolpyruvate (PEP) derived from glycolysis and erythrose-4-phosphate derived from the pentose phosphate pathway (PPP) (Carrington et al., 2018). It has been reported that the abundances of all PPP-related enzymes increased in B-deficient rape roots, indicating that the PPP was upregulated in B-deficient roots (Wang et al., 2010, 2011). Thus, it is reasonable to assume that the PPP might be induced in B-deficient orange leaves, activating the shikimate pathway. These results demonstrate that B may play a key role in the tightly coupled primary and secondary metabolic pathways, possibly the overall shift from primary to secondary metabolism due to B deficiency.

Raffinose family oligosaccharides (RFOs) are important soluble sugars in plants. They play an important role in plant development, and their accumulation in plants is correlated with stress tolerance, especially to cold stress (Bachmann et al., 1994). Raffinose is synthesized from sucrose by subsequent addition of galactose moieties provided by galactinol, which is synthesized from myo-inositol and UDP-galactose (Peterbauer and Richter, 2001). The metabolites in the raffinose pathway, such as myo-inositol, galactose, galactinol, and raffinose, showed a significant increase in B-deficient leaves relative to the control plants (Fig. 5), indicating that B deficiency enhanced the raffinose pathway. Polyols and RFOs are also important osmoprotectants that accumulate in several stress situations by supporting osmotic

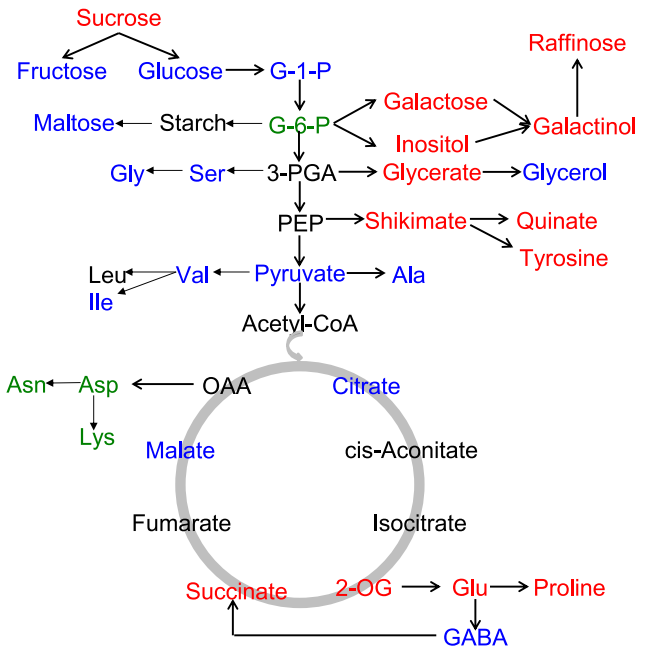


Fig. 5. Summary of metabolites and metabolic changes in leaves of navel orange plants under B deficiency. The significance of the differences between the control and B deficiency was determined by Student's *t*-test, and the metabolites with statistically significant differences ($P < 0.05$) are specially indicated. The metabolites with significantly higher levels in B deficiency than in the control are shown in a red font, and with lower levels are shown in a blue font. The metabolites displayed in a light green font had no significant changes between B deficiency and the control. The metabolites in black indicate that they are below detection levels (not detected).

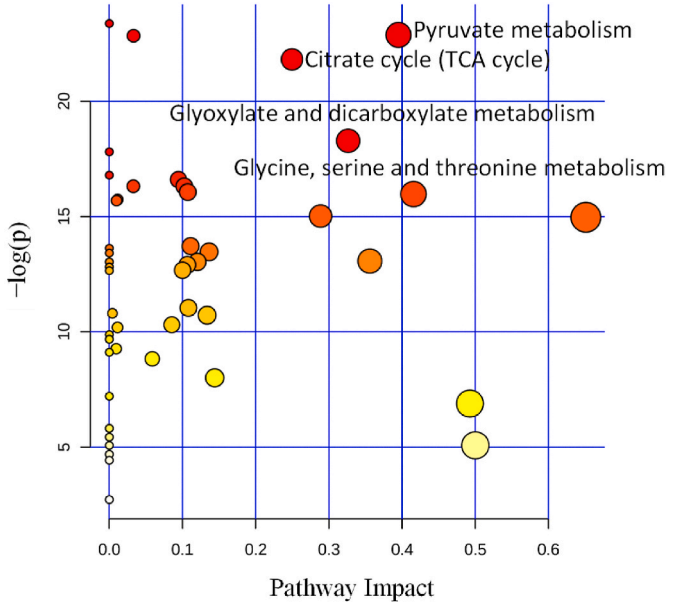


Fig. 6. Pathway enrichment analysis of the altered metabolites upon boron deficiency. The 55 well-identified altered metabolites affected by B deficiency were mapped to the biological pathways in the KEGG database, which were assigned to 47 pathways. Holistic pathway enrichment analysis applied by MetaboAnalyst 4.0 showed that these metabolites were primarily involved in pyruvate metabolism, citrate cycle (TCA cycle), glyoxylate and dicarboxylate metabolism, and glycine, serine and threonine metabolism.

adjustment (Nishizawa-Yokoi et al., 2008). Our previous results showed that B deficiency affects the water content in citrus plants (Liu et al., 2013), which could explain the increase in the myo-inositol and raffinose contents.

By contrast, and in agreement with the above explanation, the concentrations of amino acids derived from the glycolytic pathway (serine, glycine, valine, and alanine) of B-deficient leaves decreased, which may be caused by the decreased supply of C skeletons. In addition, glycine and serine both are metabolites associated with photorespiration. Chen et al. (2014) suggested that B stress (deficiency and toxicity) affected peroxisomal photorespiratory protein. The relatively rapid effects of B deficiency on proteins involved in carbohydrate and energy metabolism were also observed in rape roots (Wang et al., 2011). Indeed, the key metabolites citrate and malate of the TCA cycle decreased, similar to our previous results (Liu et al., 2015). Amino acids derived from the TCA cycle did not change or even increased relative to the control, supporting our speculation of the change in the mode of TCA under B deficiency (Liu et al., 2015).

4.2. *Exogenous citrate differentially changed metabolite profiles in plants with and without B supply*

Several possible roles have been suggested for citrate in plants, including cytosolic regulation of pH (Kania et al., 2003), the control of carbon and nitrogen metabolism (Sweetlove et al., 2010), protection against biotic and abiotic stresses (Panchal et al., 2021), and as a metabolite signaling molecule (Finkemeier et al., 2013). However, metabolic responses to citrate in citrus plants have not been identified. In this work, we investigated metabolomic profiling in navel orange leaves after the foliar application of citrate under BA and BD conditions, so as to better understand the effect of citrate on metabolism. Our results showed that exogenous foliar application of citrate changed the metabolic profiling in navel orange plants under both B supply and B deficiency conditions (Fig. 7).

Previous reports showed that citrate accumulates in plants upon biotic and abiotic stress, which can result in altered physiological processes (López-Bucio et al., 2000). It has been reported that the increased synthesis of citrate and malate is intimately related to plant tolerance to Al stress (Zhou et al., 2018). However, in our previous and the present studies, we found that the level of citrate decreased in citrus plants upon B deficiency and this decrease was a relatively early effect (before the occurrence of visible symptoms) of B deficiency. Endogenous organic acids are the source of both carbon skeletons and energy for cells and are linked to various metabolic pathways in different compartments, making them ideally placed to function as reporter molecules of metabolic states (Sweetlove et al., 2010). Therefore, it is possible that the changes in metabolic profiling due to B deficiency, such as the promoted shikimate pathway and raffinose pathway, were caused partially by the decreased level of citrate (Fig. 5). Indeed, the level of endogenous citrate was increased in citrate-treated orange plants. This induced the citrate into the TCA cycle, which could restore central carbon metabolism then inhibited the shikimate pathway and raffinose pathway (Fig. 9). Based on the results, the role of citrate on metabolite changes under B deficiency conditions is mainly reflected in its metabolic recovery effect.

However, the effect of citrate on the metabolic profiles of leaves under normal conditions is very different from that under B deficiency conditions. For example, application of citrate led to a significant increase and decrease in tyrosine level in orange leaves with and without B supply, respectively (Fig. 7). Similarly, exogenous citrate also increased the level of tyrosine in *Arabidopsis* leaves under normal conditions (Finkemeier et al., 2013). Citrate has been identified as an important player in metabolite signaling in plant cells, besides its conserved function as a TCA cycle intermediate (Finkemeier et al., 2013). It has been reported that citrate has a strong effect on transcript abundances in *Arabidopsis* leaves, and the regulation of transcript abundance by altered citrate level is time and concentration dependent (Finkemeier et al.,

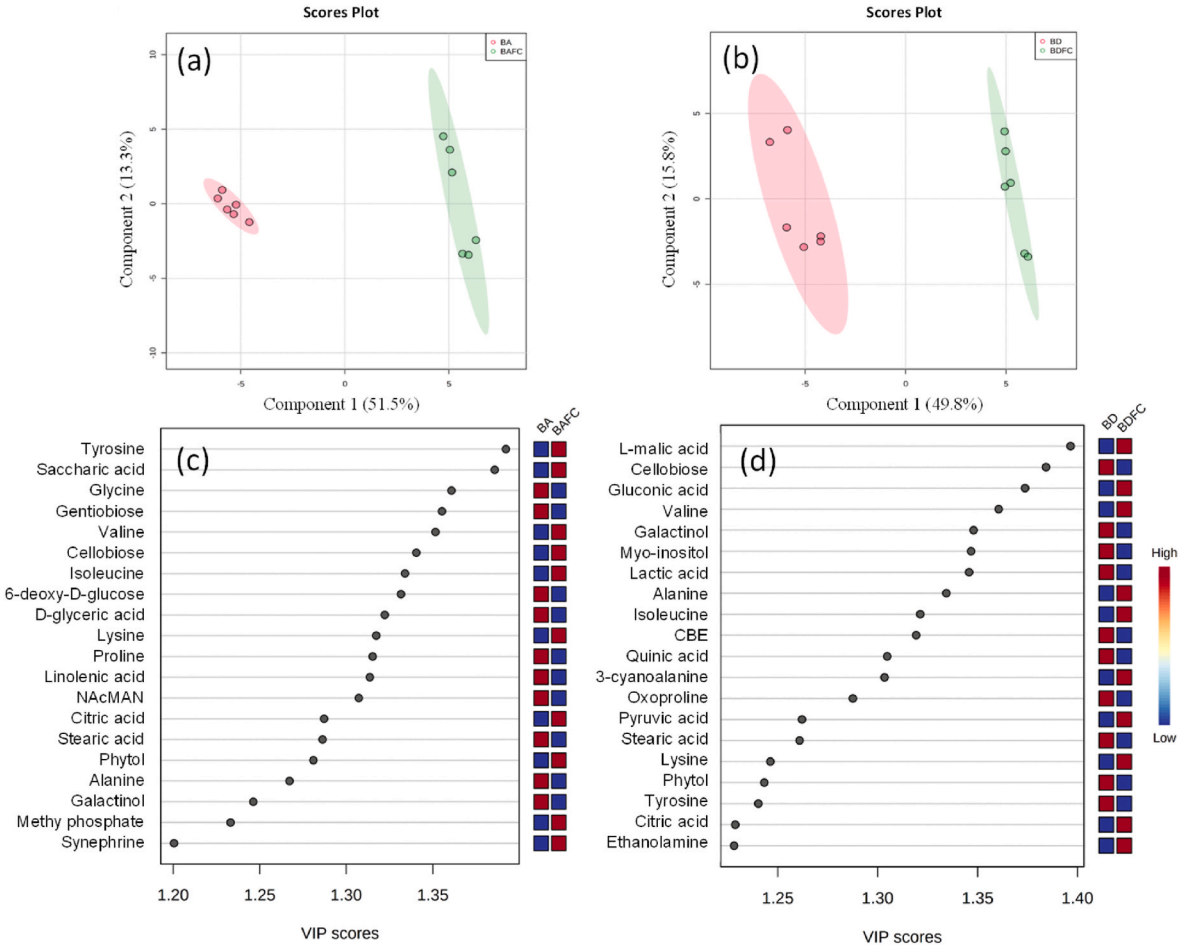


Fig. 7. Partial least square-discriminant analysis (PLS-DA) of leaf metabolite profiles between BAFC and BA (a and c) and BDFC and BD (b and d). (a and b) PLS-DA carried out using the 64 identified metabolites. (c and d) Variable importance in projection (VIP) scores of the PLS-DA. The figure shows the top 20 metabolites contributing to component 1. These analyses were carried out using the MetaboAnalyst platform. (NAcMAN: N-Acetyl-beta-D-mannosamine; CBE: conduritol b epoxide).

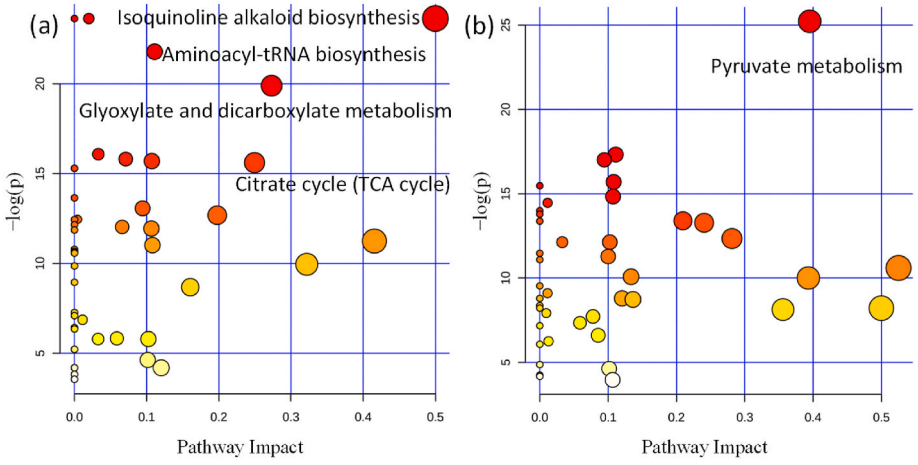


Fig. 8. Pathway enrichment analysis of the altered metabolites upon foliar citrate under the boron adequate condition (a) and boron deficiency condition (b).

2013). This may explain why foliar application of citrate differentially changed metabolite profiles (including tyrosine) in leaves of orange plants with and without B supply (Fig. 7), since B deficiency affected the level of citrate in leaves (Fig. 5). Taken together, we speculate that citrate affected the metabolic profiling of leaves as a metabolite signaling molecule, which regulated transcript abundance in navel orange plants under normal conditions. Based on the above analysis, these results indicate that there are different responsive and regulatory mechanisms with respect to the additions of exogenous citrate in navel orange leaves under B adequate and deficient conditions.

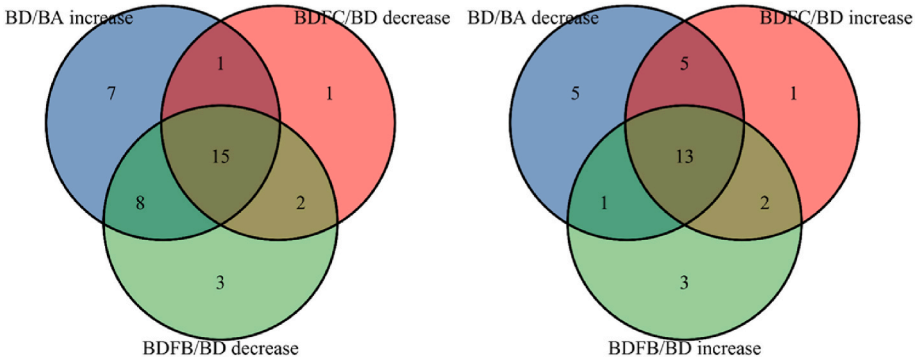


Fig. 9. Venn diagram showing the numbers and metabolites that overlapped among B deficiency and foliar application of citrate or borate under B deficiency conditions.

4.3. The metabolome response to citrate is similar to borate treatment under BD conditions

Several hypotheses have been proposed for the mechanism by which exogenous citrate mitigates stress in plants. It was suggested that citrate increases heat tolerance by enchaining of photosynthesis and maintaining membrane stability (Hu et al., 2016). However, it was also suggested that citrate mitigated plant heavy metal stress by maintaining cellular redox homeostasis (Massoud et al., 2019). In addition, the increases in citrate can up-regulate the TCA cycle to provide more energy to support the costly defense responses to biotic stress (Igamberdiev and Eprintsev, 2016). It is also believed that exogenous citrate can improve plant tolerance to stress by induction of the antioxidant enzymes such as catalase and peroxidase (Sun and Hong, 2011). In contrast to the increases in citrate contents in other abiotic-stressed plants (Canarini et al., 2019), B deficiency decreased the concentrations of citrate in orange leaves. Furthermore, exogenous B (borate) had the ability to increase the internal citrate concentrations in the leaves of B-deficient orange plants. In addition, exogenous citrate also increased the accumulation of citrate in the leaves of B-deficient plants in vivo. These results indicated that B may be involved in citrate metabolism.

In the present study, contents of most of the identified metabolites were significantly changed after BD treatment, while foliar borate treatment reversed the effect (Fig. 9). Consistent with this, B resupply can rapidly reverse the B starvation-induced changes in metabolism and increase the number and length of lateral roots in citrus plants (Wu et al., 2019). Interestingly, foliar citrate also reversed the concentrations of many metabolites that were changed by BD treatment. There were 28 metabolites with similar recovery tendency between the BDFB and BDFC leaves relative to the BD leaves. Furthermore, the cluster analysis revealed a high similarity of the metabolic response to citrate and borate in BD plants. Although the exact mechanism of citrate treatment response under B deficiency is difficult to infer, this study clearly demonstrates the metabolic recovery effect of citrate for B-deficient stress, very similar to the B resupply effect.

According to the results of this work and those recently published by our group (Zhou et al., 2022), we propose a possible model to explain metabolic regulation by foliar application of citrate or borate in navel orange plants under B deficiency conditions (Fig. 10). Boron deficiency resulted in decreased citrate and pyruvate levels in navel orange leaves, which destroyed central carbon metabolism and led to the carbon flux shifted towards the shikimate pathway and raffinose pathway. After exogenous application of borate or citrate, the endogenous citrate concentrations increased. This induced the conversion of citrate to malate and then to pyruvate (López-Bucio et al., 2000; Sweetlove et al., 2010), which restored central carbon metabolism and inhibited the shikimate pathway and raffinose pathway. Taken together with the pathway enrichment analysis results, it is possible that citrate regulates pyruvate metabolism under B deficiency to restore metabolic homeostasis.

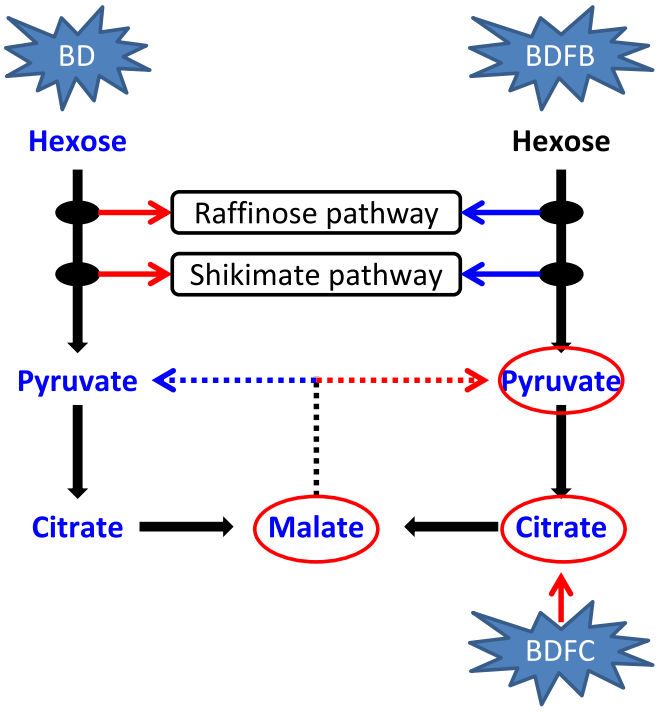


Fig. 10. Proposed model of metabolic regulation by foliar application of citrate or borate in navel orange plants under boron deficiency (BD) conditions. The metabolites with significantly lower levels in BD than in the control are shown in a blue font, and with increased levels after foliar application of citrate or borate under BD conditions are shown in a red oval. The blue and red arrows represent decreases and increases, respectively.

Author statement

Guidong Liu and Gaofeng Zhou conceived the project and designed the experiments. Guidong Liu performed the metabolomics analysis. Tian Jin and Yuemei Xu performed the determination of boron concentration. Guan Guan performed the statistics analysis. Guidong Liu and Gaofeng Zhou prepared all the figures and wrote the manuscript assisted by Fengxian Yao. All the authors read and approved the final manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.plaphy.2022.09.033.

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